

# Po-Wei (Brian) Chen

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## EDUCATION

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**Soochow University**, Taipei, Taiwan

B.B.A. in Financial Engineering and Actuarial Mathematics

Sept. 2022 – June 2025

- Graduated one year early (met criteria of full credit completion and top 10% class rank each semester)
- Cumulative GPA 4.0/4.0; Average 95.2/100; Class Rank 1<sup>st</sup>/53

## WORK EXPERIENCE

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**Research Assistant (Full-time)**

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Aug. 2025 – present

PI: Prof. Hsin-Cheng Huang, Distinguished Research Fellow

- Conduct research on probabilistic generative models to improve predictive inference in limited-data settings, and currently explore conformal prediction for spatial and spatio-temporal data.
- Managed administrative tasks and assisted with operations for international conferences.

**High School Math Tutor**

Aug. 2024 – June 2025

Taiwan TACT Education (Private Tutoring Center), Minsheng Branch, Taipei, Taiwan

- Taught high school mathematics and prepared students for university entrance examinations.

## HONORS & AWARDS

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**ISI-ISM-ISSAS Joint Conference Poster Competition Award (2026)**

Title: Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors.

**Outstanding Graduating Student (2025)**

Soochow University, Taipei, Taiwan

**Academic Excellence Award (2022–2025; six-time recipient; awarded to top three of 50+ students each semester)**

Soochow University, Taipei, Taiwan

## RESEARCH PROJECTS

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**Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models**

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

- Augmented diffusion-based generative models with a Bayesian framework of catalytic priors that leverage simpler statistical structures to guide learning in limited-data settings.
- **My contribution:** Conducted simulation experiments to evaluate improvements in predictive accuracy and uncertainty quantification across different data settings.

**Empirical Asset Pricing via Machine Learning for Cryptocurrency**

Soochow University, Taipei, Taiwan

- Proposed a portfolio construction framework for cryptocurrency markets using machine learning predictions, linking predicted returns to financial characteristics through cross-sectional rankings.
- **My contribution:** Contributed to framework development, implemented and evaluated multiple machine learning models, and conducted empirical analyses comparing the proposed framework with benchmark strategies.

## CONFERENCE PRESENTATIONS

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- **“Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models”** 14<sup>th</sup> IASC-ARS & SoAI Joint Conference, Chiang Mai University, Chiang Mai, Thailand. Dec. 2026 [Accepted for Oral Presentation]
- **“Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors”** ISI-ISM-ISSAS Joint Conference, Academia Sinica, Taipei, Taiwan. Feb. 2026 [Poster]
- **“Empirical Asset Pricing via Machine Learning for Cryptocurrency”** Annual Conference and International Symposium of the Taiwan Risk and Insurance Association, Tamkang University, New Taipei, Taiwan. Dec. 2025 [Oral]

## COURSEWORK & CERTIFICATIONS

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### Selected Coursework

Soochow University, Taipei, Taiwan

- **Mathematics & Analysis:** Advanced Calculus (Introduction to Analysis), Calculus I–II, Linear Algebra I–II
- **Statistics:** Probability and Statistics I–II, Statistical Inference, Advanced Statistics, Regression Analysis, Applied Statistics
- **Computational & Optimization Methods:** Numerical Analysis, Operations Research, Artificial Intelligence in Finance, Computer Programming I–II, C++ Programming Language
- **Actuarial & Financial Mathematics:** Fundamentals of Actuarial Mathematics, Mathematics of Compound Interest, Financial Calculation

### Open Online Courses

- Python for Everybody Specialization, University of Michigan (via Coursera)
- Programming for Business Computing in Python (1), National Taiwan University (via Coursera)

### Certification

- Actuarial Exam P (Probability), Society of Actuaries (SOA) — Passed

## SKILLS & TECHNICAL

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**Programming:** Python, R (R Markdown, Shiny), C++, SQL

**Machine Learning Packages:** PyTorch, TensorFlow, scikit-learn

## EXTRACURRICULAR ACTIVITIES

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### Statistics Cup Basketball Tournament Champions

Nov. 2025

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

### Vice Captain of Basketball Team

2024 – 2025

Department of Financial Engineering and Actuarial Mathematics, Soochow University, Taipei, Taiwan

- Led the team to six championships in inter-departmental competitions.